

MA 301

Test 3

Fall - 2013

Problem	points	out of
1		10
2		10
3		12
Score		32

Name: SOLUTIONS

HR: _____

Signature: _____

Formula Sheet

Fourier Series: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L}$ where $d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi n x/L} dx$

if $f(x)$ is even then $d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx$

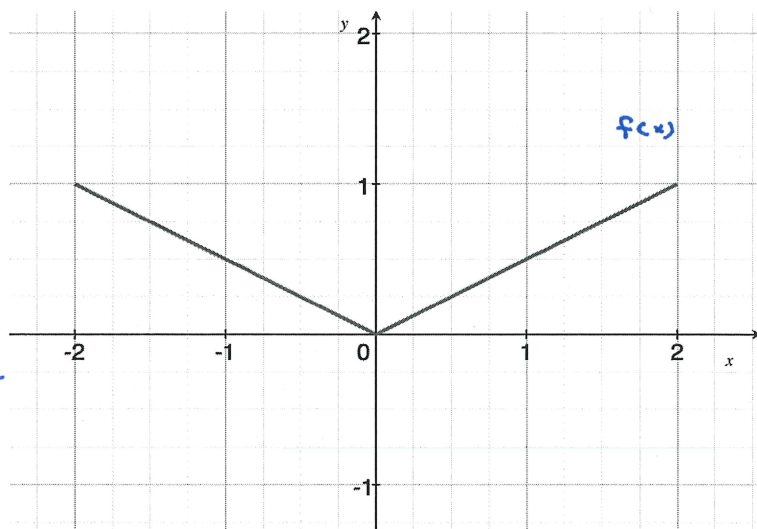
if $f(x)$ is odd then $d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx$

Problem 1: Determine the Fourier Series (don't just find coefficients d_n) for the function depicted below. Assume it periodically extends along the x -axis.

$$f(x) = \begin{cases} -\frac{1}{2}x & x \leq 0 \\ \frac{1}{2}x & x \geq 0 \end{cases}$$

$f(x)$ is even

$$f(x) = f(x+4) \Rightarrow P = 4 = 2L \\ \Rightarrow L = 2$$



$$d_0 = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} \left(\frac{1}{2}(2)(1) \right) = \frac{1}{2}$$

$$d_n = \frac{1}{2} \int_0^2 \frac{1}{2} x \cos\left(\frac{\pi}{2}nx\right) dx$$

$$u = x \quad v = \frac{2}{\pi n} \sin\left(\frac{\pi}{2}nx\right) \\ du = dx \quad dv = \cos\left(\frac{\pi}{2}nx\right) dx$$

$$= \frac{1}{4} \left[\frac{2}{\pi n} x \sin\left(\frac{\pi}{2}nx\right) - \frac{2}{\pi n} \int \sin\left(\frac{\pi}{2}nx\right) dx \right]_0^2$$

$$= \frac{1}{4} \left[\frac{2}{\pi n} x \sin\left(\frac{\pi}{2}nx\right) + \left(\frac{2}{\pi n}\right)^2 \cos\left(\frac{\pi}{2}nx\right) \right]_0^2$$

$$= \frac{1}{4} \left[\frac{2}{\pi n} (2) \sin(\pi n) + \left(\frac{2}{\pi n}\right)^2 \cos(\pi n) - 0 - \left(\frac{2}{\pi n}\right)^2 \cos(0) \right]$$

$$= \frac{1}{4} \left[\left(\frac{2}{\pi n}\right)^2 (-1)^n - \left(\frac{2}{\pi n}\right)^2 \right] = \frac{1}{\pi^2 n^2} [(-1)^n - 1]$$

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\frac{\pi}{2}nx}$$

where

$$d_0 = \frac{1}{2}$$

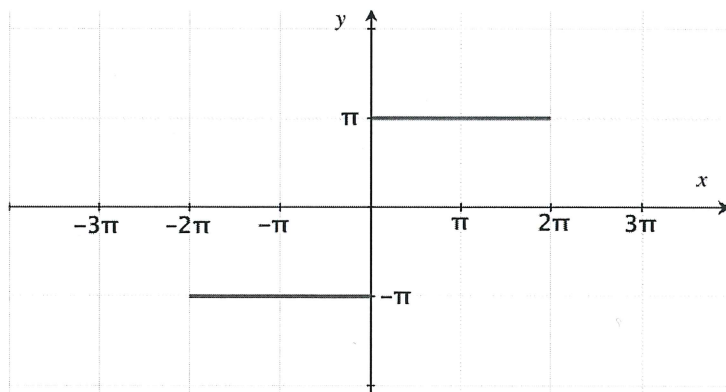
$$d_n = \frac{1}{\pi^2 n^2} [(-1)^n - 1]$$

Problem 2: Determine the Fourier Series (don't just find coefficients d_n) for the function depicted below. Assume it periodically extends along the x -axis.

$$f(x) = \begin{cases} -\pi & x \in (-2\pi, 0] \\ \pi & x \in (0, 2\pi) \end{cases}$$

$f(x)$ is odd

$$f(x) = f(x+4\pi) \Rightarrow L = 2\pi$$



$$d_0 = 0$$

$$d_n = \frac{-i}{2\pi} \int_0^{2\pi} \pi \sin\left(\frac{nx}{2}\right) dx$$

$$= \frac{-i}{2} \left[-\left(\frac{2}{n}\right) \cos\left(\frac{nx}{2}\right) \right]_0^{2\pi}$$

$$= + \frac{i}{n} (\cos(\pi n) - \cos(0))$$

$$= + \frac{i}{n} ((-1)^n - 1)$$

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{\frac{inx}{2}}$$

where

$$d_0 = 0$$

$$d_n = \frac{i}{n} ((-1)^n - 1)$$

Problem 3: For each function $f(x)$ below with specified period, Setup (DO NOT Compute) the integral (or sum of integrals) needed to compute the Fourier coefficients. $f(x)$ should not be in your answer.

(a) $f(x) = \begin{cases} 0 & -2 \leq x \leq 0 \\ 1 & 0 \leq x \leq 1 \\ 0 & 1 \leq x \leq 2 \end{cases}$ where $f(x) = f(x+4)$ $L=2$

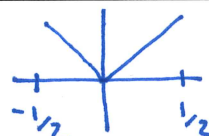


EVEN

$$d_0 = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} \int_0^1 1 dx$$

$$d_n = \frac{1}{2} \int_0^2 f(x) \cos\left(\frac{\pi}{2} nx\right) dx = \frac{1}{2} \int_0^1 \cos\left(\frac{\pi}{2} nx\right) dx$$

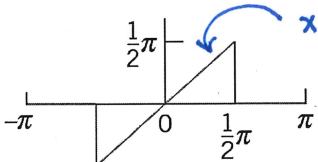
(b) $f(x) = |x|$ where $f(x) = f(x+1)$ $L = \frac{1}{2}$



EVEN

$$d_0 = 2 \int_0^{1/2} f(x) dx = 2 \int_0^{1/2} x dx$$

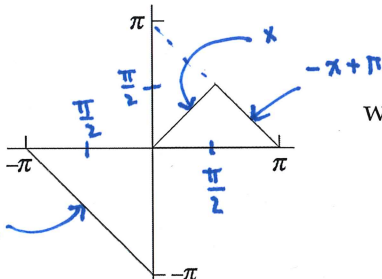
$$d_n = 2 \int_0^{1/2} f(x) \cos(2\pi nx) dx = 2 \int_0^{1/2} x \cos(2\pi nx) dx$$

(c)  where $f(x) = f(x+2\pi)$ $L = \pi$

ODD

$$d_0 = 0$$

$$d_n = \frac{-i}{\pi} \int_0^{\pi} f(x) \sin(nx) dx = -\frac{i}{\pi} \int_0^{\pi} x \sin(nx) dx$$

(d)  where $f(x) = f(x+2\pi)$ $L = \pi$

$$d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-x-\pi) dx + \int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi} (-x+\pi) dx \right]$$

$$d_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-x-\pi) e^{-inx} dx + \int_0^{\pi/2} x e^{-inx} dx + \int_{\pi/2}^{\pi} (-x+\pi) e^{-inx} dx \right]$$